Unit 3

Screw Thread & Gear Measurement, CMM

(Handwritten Notes)

Syllabus Topics Covered

1. Elements of screw threads; errors in screw threads
2. Measurement of external and internal threads
3. Screw thread gauges
4. Gear tooth terminology
5. Measurement of gear elements: run-out, lead, pitch, backlash, profile, pressure angle
6. Tooth thickness, diameter, constant chord and base tangent methods
7. Coordinate Measuring Machine (CMM): construction and features

1. Screw Thread Measurement

1.1 Elements (terminology) of a screw thread

Basic profile of an ISO metric (60 degree) thread

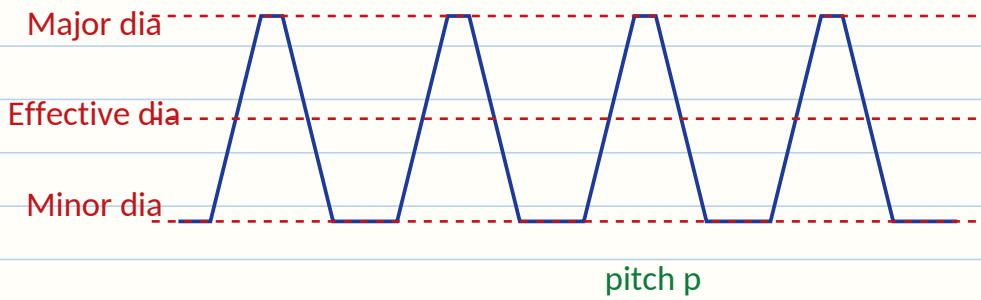


Fig 3.1: Thread profile with major, effective (pitch) and minor diameters

Term	Definition
Major (outside) diameter	largest diameter of a thread (crest of external, root of internal); **nominal size**

Minor (core / root) diameter	smallest diameter of the thread (root of external, crest of internal)
Pitch (effective / simple effective) diameter	diameter of an **imaginary co-axial cylinder** which cuts the thread so that the **width of the groove = width of the thread** (at that cylinder)
Pitch (p)	distance from a point on one thread to the corresponding point on the **next thread** , **parallel to the axis**
Lead (L)	axial distance **advanced in one revolution** : **L = n x p** (n = number of starts)
Flank angle / thread angle	angle between flank and the perpendicular to the axis / included angle between the two flanks (60 deg metric; 55 deg Whitworth)
Crest & root	top surface joining two flanks / bottom surface between two flanks
Depth of thread	distance between crest and root measured **perpendicular to the axis**
Helix angle	angle made by the helix of thread at the pitch cylinder: **$\tan \psi = L / (\pi d_2)$**
Flank	straight surface connecting crest and root

Thread form	Angle	Use
ISO metric	60 degree (flat crest)	fasteners, general
Whitworth (BSW)	55 degree (rounded)	British pipe & fasteners
ACME	29 degree	lead screws, power transmission
Square	0 degree flanks	vices, jacks (high load, low friction)
Buttress	45 / 7 degree	one-direction heavy load (press, jacks)

1.2 Errors in screw threads

1. **Pitch errors** : (a) **Progressive** - pitch uniformly longer or shorter than nominal over the length (error in lead screw of lathe; **tool-feed** or **expansion** errors). (b) **Periodic** - repeats at regular intervals per revolution (eccentric mounting of blank / **gear errors** in the lead-screw drive). (c) **Drunkenness** - the thread form

is a **helix with a periodic variation** : axial section looks like straight but the **thread curve is not perfect helix** (cutting tool inclined). (d) **Irregular** - due to machine, tool wear, material variations.

2. **Major / minor diameter errors** : cause interference with mating parts at the root/crest.
3. **Effective diameter errors** - most critical: affect the fit (due to wrong depth of cut).
4. **Flank angle errors** : **angle error** (angle of both flanks differ from nominal) or **unequal flank angles / bisector not perpendicular to axis** (tool setting).
5. **Form errors** : wrong crest/root radius, **rounded** or irregular profile.

Effect of pitch / angle error

A pitch error or flank-angle error **virtually increases the effective diameter** (of an external thread):

- **Virtual effective diameter = measured effective diameter + (compensation for pitch error P + flank angle error)**
- Pitch error compensation: **1.732 |ΔP|** , Angle compensation: **0.0003 p (Δα in minutes)** (metric 60 degree).

1.3 Measurement of external screw threads

(a) Major diameter

- **Bench micrometer** (or ordinary **micrometer** with flat anvils) measures the outside diameter over the crests; **bench micrometer** with **standard cylinder** gives comparison with a **setting cylinder** of equal size, with constant pressure (fiducial indicator). Accuracy 0.002-0.005 mm.

(b) Minor diameter

- Measured with a **micrometer** fitted with **tapered V anvils / small radius pieces (prism pieces)** touching the root of the thread. Or **toolmaker's microscope** .

(c) Effective (pitch) diameter

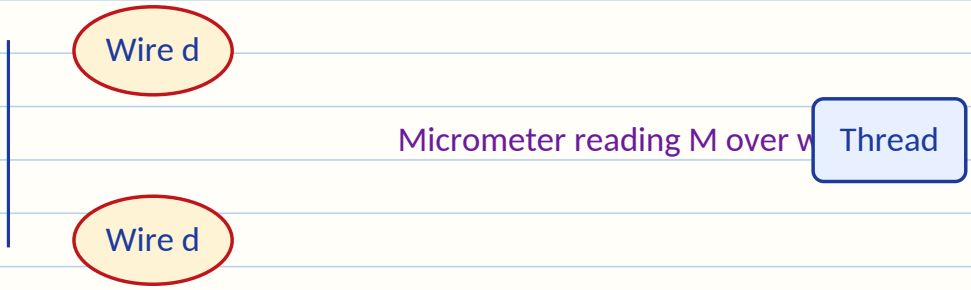
1. Screw thread micrometer - the anvil is a **V** (60 deg) and spindle a **cone** that fit the thread flanks; reading gives the effective diameter directly (error: pitch error of the micrometer; need set for each pitch range).

2. Two-wire method : two **accurately ground wires of same diameter d** are placed in **opposite grooves** and **measured over the wires by a micrometer** (M). Third wire is not used.

Two / Three wire formulae (metric 60 degree thread)

Effective diameter E = M - P where **P = d (1 + cosec (α/2)) - (p/2) cot (α/2)**

- For **metric** (α = 60): **P = 3d - 0.866 p** ; For **Whitworth** (α = 55): P = 3.166 d - 0.9605 p.
- **Best wire size** (touches the flanks on the **pitch line**): **d = p / (2 cos (α/2))** = **0.5774 p** (metric) = 0.5637 p (Whitworth).
- **Over the wires: M = E + 3d - 0.866 p** (metric).



- **Example** : M10 x 1.5 thread, best wire d = 0.5774 x 1.5 = **0.866 mm** ; E = 9.026 mm -> M = 9.026 + 3 x 0.866 - 0.866 x 1.5 = 9.026 + 2.598 - 1.299 = **10.325 mm** .

3. Three-wire method : **three wires of same diameter** - two on one side and one on the opposite side; the micrometer is placed over the **single wire and the pair** - **more accurate** than two-wire as it **aligns the axis** of the micrometer perpendicular to the thread axis, and cancels the effect of inclination.

4. Floating carriage micrometer (Diameter measuring machine) : the thread is mounted **between centres** ; carriage (with a micrometer and **fiducial indicator**) floats freely on ball slides. A **setting cylinder** of known diameter and wires are used: **E = (cylinder effective size) + (difference of readings: thread - cylinder)** (comparison method). Used also to measure **major & minor diameters** and **pitch** .

(d) Pitch measurement

- **Pitch measuring machine** : the thread is **mounted between centres** ; a **stylus with a ball** (or a conical point) is brought into the successive grooves and the carriage advance is read with a **micrometer** - **cumulative pitch error** and **progressive** or **periodic** errors can be seen. **Screw pitch gauge** (set of leaves) gives **quick comparison** of the pitch.
- **Toolmaker's microscope** (also **optical projector**): pitch measurement along axis

with the cross hair at the corresponding points on the successive threads.

(e) Thread flank angle & form

- **Toolmaker's microscope** (angular scale in the eyepiece, crosshair aligned with flank) - angle by rotating the eyepiece, **profile projector** with **master chart (overlay)** ; **optical comparator** (shadow method) - simple and fast.

2. Measurement of Internal Threads

More **difficult** than external threads because the thread is **inside** a hole, and a standard micrometer cannot reach.

- **Major & minor diameters** : **taper parallel (plug) setting** with a micrometer; **internal micrometer** , **telescopic gauge** , **ball-end** inside caliper; the **minor diameter** is checked by **GO / NO GO plain plug gauge** , and the **major** by thread plug.
- **Effective diameter** : **with taper parallels and rollers (balls)** - a pair of **taper parallels** with **three rollers** etc., so the reading over the rollers is taken with a micrometer; alternatively **ball method** (balls of best size inserted in the groove and **depth micrometer / height gauge** measures the distance across).
- **Casts (impression)** : take a **cast of the internal thread** (plaster / metal / resin cast) and measure the cast as an external thread.
- **Pitch & angle** : **toolmaker's microscope** after **sectioning** , or by **impressions** ; **optical** (using a prism-and-microscope arrangement) via **bore scope** .
- **Thread plug gauges** (GO / NO GO) are the most **practical** shop floor inspection method.

Screw thread gauges

Thread gauges (limit gauges) check the **conformity** of thread within limits **quickly** , according to **Taylor's principle** .

Gauge	Checks	Details
Thread plug gauge	internal thread (nut)	**GO** : **full form** , long (engages full length) - checks **effective dia + major + minor** . **NO GO** : **truncated flanks** , **short** (2-3 threads) - checks **effective dia only** ; **minor diameter** checked by plain plug
Thread ring gauge	external thread (bolt)	GO ring: full form; **NO GO ring** : truncated thread, shorter. **Adjustable ring gauges** for wear

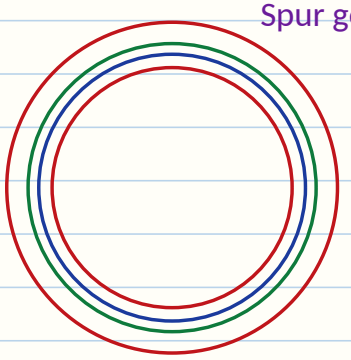
Caliper (snap) thread gauge	external thread	two pairs of rolls / anvils; GO rolls full form; NO GO rolls flank contact only
Screw pitch gauge	pitch (identification)	set of **steel leaves** with thread forms; select the leaf that fits
Setting plugs / check plugs	inspection of ring gauges	**setting plug** to set adjustable gauges; **check plug** to confirm wear of the GO gauge

- **NO GO is truncated** (flank contact only at the pitch diameter) so a **fault in major/minor diameter** doesn't make it pass/reject wrongly; thread NO GO gauge **must not enter more than 2 turns** .
- **Tolerance classes (IS 4218)** : 6H (internal medium), 6g (external medium); **fine / medium / coarse** classes (4H5H, 6H, 7H; 4h, 6g, 8g). **Designation** : **M10 x 1.5 - 6H** .

3. Gear Measurement

3.1 Gear tooth terminology (spur gear)

Spur gear tooth terminology



Addendum circle

Pitch circle

Dedendum (root) circle

Base circle

Term	Meaning / formula
Pitch circle (diameter D)	imaginary circle that **rolls without slipping** with the pitch circle of the mating gear
Module (m)	**m = D / N** (mm) - pitch diameter divided by number of teeth (metric standard)
Diametral pitch (P)	**P = N / D** (teeth per inch); **m x P = 25.4**
Circular pitch (p_c)	distance between corresponding points of adjacent teeth along the pitch circle: **p_c = π m = π D / N**
Addendum (a)	radial height of tooth **above** the pitch circle (= **1 m**); **Dedendum (d)** = below the pitch circle (= **1.25 m**)

Clearance (c)	dedendum - addendum = **0.25 m**
Pressure angle (φ)	angle between the **line of action** (common normal) and the **common tangent** to pitch circles: **14.5 or 20 degree** standard
Base circle	circle from which the involute is generated: **$D_b = D \cos \varphi$** ; **base pitch = $\pi m \cos \varphi$**
Tooth thickness (t)	arc thickness at pitch circle = **$\pi m / 2$ = 1.5708 m**
Backlash	clearance between non-working flanks of mating teeth along the pitch circle
Face width, lead (helical), whole depth	width of tooth parallel to axis; whole depth = **2.25 m** ; **outside dia = m (N + 2)**

Errors in gears

- **Profile (involute) error** , **pitch error** (**single/adjacent** , **cumulative**), **lead error** (helical), **radial run-out** (eccentricity), **tooth thickness** error, **backlash** , **alignment (parallelism / skew) error** , **composite error** (**total & tooth-to-tooth**), **surface finish** .
- **Causes** : errors in the **gear cutting machine** (**hob** , indexing/worm gear), **improper mounting** of blank, **tool wear** , **heat-treatment distortion** and **misalignment** .

4. Measurement of Gear Elements

4.1 Run-out (radial eccentricity)

Run-out = variation in the **radial position** of the **pitch line** with respect to the **axis of rotation** , due to **eccentric mounting** or an **inaccurate** gear blank.

- **Method** : gear is **mounted between centres** (or on an **accurate mandrel**); a **dial indicator** (with a **ball or cylindrical probe / roller** of suitable diameter) is placed in **successive tooth spaces** - **the pointer variation (Max - Min)** = run-out. Plotting gives **eccentricity** .

4.2 Pitch measurement

- **Pitch error** = difference between the actual and theoretical pitch. **Adjacent (single) pitch error** and **cumulative pitch error** (sum over a sector or full circle).
- **Pitch measuring instrument** (**Gear pitch comparator**): has a **fixed finger** and **adjustable finger** touching two successive teeth at the pitch circle; a **dial indicator** shows the **deviation** from the **set master pitch** (set using slip gauges). Both **base-pitch** and **circular-pitch** measurement.

- **Base pitch (normal pitch) measuring : base-pitch gauge (Ludwig type)** with **two fixed contacts (pre-set)** touching adjacent teeth flanks **tangentially** along the **line of action : base pitch $p_b = \pi m \cos \varphi$** (module 4, 20 degree: 11.81 mm).
- **Using optical dividing head / indexing table** with a **dial indicator - each tooth gap** is checked at successive angular positions (**360/N**).

4.3 Lead measurement (helical gears)

- **Lead** is the axial distance through which a **helical tooth** advances in one turn. **Lead checking instrument** : gear on **centres** with **a probe contacting the tooth flank** ; **rotate the gear** while the **probe moves axially (helical guide / sine bar guide)** at the **set lead** ; the **dial indicator** shows **lead deviation** across the face width.
- For **spur gears** the **lead error = parallelism error** of tooth traces to the axis, checked by dial indicator moving parallel to the axis.

4.4 Backlash

- **Backlash = max. tooth space - min. tooth thickness** ; **needed for lubrication & thermal expansion** but too much -> noise & impact; **measured** by **fixing one gear** , **rocking the other** with a **dial indicator** at the **pitch circle** : (reading) = backlash. Or **feeler gauge** between non-driving flanks on assembly; **centre-distance** variation method.

4.5 Profile (involute) measurement

- **Involute profile** tester: **base disc (cylinder)** of **diameter = $D_b = D \cos \varphi$** mounted **co-axially with the gear** ; a **straight edge (ruler)** is held in contact with the base disc and **rolls without slip** ; a **stylus** touching the tooth flank moves by the **dial indicator - deviation from the true involute** = reading **(Hilger & Watts / Zeiss / Klingelnberg machines)** .

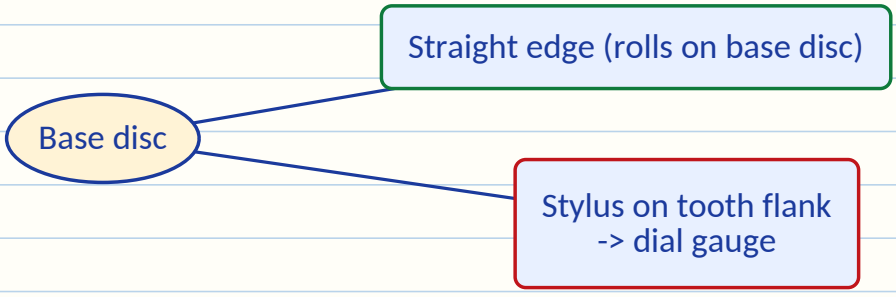


Fig 3.2: Principle of involute profile measurement (generating by rolling straight edge)

- **Optical profile projector (10x - 50x)** overlays the **magnified tooth profile** on a **drawn master chart** ; **toolmaker's microscope** ; **CMM (gear measurement**

software).

4.6 Pressure angle

- Gear pressure angle can be found from the base circle : $\cos \varphi = D_b / D$ or base pitch : $\cos \varphi = p_b / (\pi m)$. Measure base pitch with the base-pitch gauge and compare with the module (circular pitch). Also by profile projector with protractor reticle on the tooth flank at the pitch point.

4.7 Diameter of gear

- Outside (addendum) diameter : vernier caliper / micrometer (for even teeth directly; odd teeth need correction D_o = (reading) / cos (90/N)). Pitch circle diameter (PCD) : from measurement over rollers (pins) of diameter d placed in opposite tooth spaces : $M = D_b / \cos \varphi_1 + d$ (even teeth) - pin diameter (best size = 1.728 m for 20 degree); root diameter : micrometer across rollers at the root.

4.8 Tooth thickness - gear tooth vernier caliper

Tooth thickness is measured as chordal thickness (straight line between two points on the pitch circle), not arc thickness , with a gear tooth vernier caliper (two vernier scales : one horizontal for chord width w , vertical for the depth h (chordal addendum)).

Gear tooth vernier:

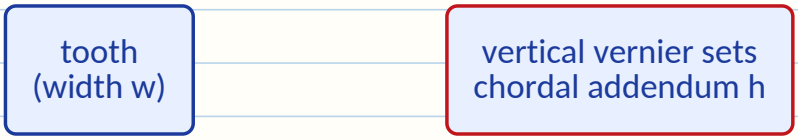


Fig 3.3: Gear-tooth vernier caliper

Chordal thickness and chordal addendum (spur gear, N teeth, module m)

Chordal thickness $w = N m \sin (90 / N)$

Chordal addendum $h = m + (N m / 2) [1 - \cos (90 / N)]$

- Arc tooth thickness = $\pi m / 2$ on pitch circle; chord is shorter. Vernier vertical scale set to h ; horizontal reads w - compare with computed.

Example : m = 4, N = 30: w = 30 x 4 x sin 3 degree = 120 x 0.05234 = 6.280 mm ; h = 4 + (120 / 2)(1 - cos 3 degree) = 4 + 60 x 0.001370 = 4.082 mm .

4.9 Constant chord method

- **Constant chord** = the **chord joining the points where the tooth flanks touch a standard rack (of the same module & pressure angle)** ; its length is **constant for all gears of the same module** and pressure angle - **independent of number of teeth** .

Constant chord formulae

$t_c = (\pi m / 2) \cos^2 \varphi$ (constant chord)

$h_c = m - (\pi m / 8) \sin 2\varphi$ (height of chord from tooth tip)

- **Example** m = 4, φ = 20 deg: t_c = 6.283 x 0.8830 = **5.548 mm** ; h_c = 4 - 1.5708 x 0.6428 = **2.990 mm** .

- **Advantages** : **no need to know the number of teeth** ; **no need to correct for each gear** ; suitable for **interchangeable gears** . Disadvantage: **depends on the vernier caliper depth setting accuracy** , needs **pressure angle** known.

4.10 Base tangent method

- **Span measurement** : measure the distance **W** across **k teeth** with a **disc micrometer / vernier (or flange) micrometer** whose anvils touch **opposite flanks** , **tangent to the base circle** (the common tangent = **base tangent length**). It is **independent of the gear axis** (no run-out effect) and **depends only on the base pitch** .

Base tangent length

$W = m \cos \varphi [\, \pi (k - 0.5) + N \, \text{inv}\, \varphi \,]$ with $\text{inv}\, \varphi = \tan \varphi - \varphi$ (rad) (inv 20 deg = 0.014904)

- Number of teeth to span: $k = N \varphi / 180 + 0.5$ (nearest integer); $W = (k - 1) p_b + s_b$ (base thickness).

Example : m = 4, N = 30, φ = 20 deg: k = 30 x 20 / 180 + 0.5 = 3.83 -> **4** ; W = 4 x 0.93969 [π (3.5) + 30 x 0.014904] = 3.75877 x (10.9956 + 0.4471) = **43.01 mm** .

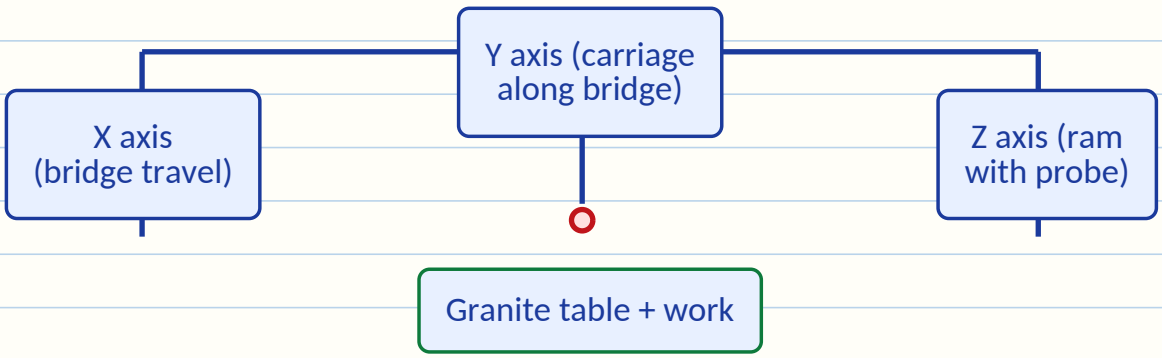
4.11 Composite error - Parkinson gear tester

- **Composite action test** : the gear under test is **meshed with a master gear** with **no backlash (double-flank contact)** under **spring pressure** ; one gear is on a **fixed slide** , the other on a **floating (sliding) carriage** ; **centre-distance variation** during **one revolution** is **recorded on a chart / dial** : **total composite error** = max variation; **tooth-to-tooth composite error** = max change per tooth. **Quick** and checks **overall quality** (run-out + profile + pitch).

5. Coordinate Measuring Machine (CMM)

A **Coordinate Measuring Machine (CMM)** is a **mechanical system** that moves a **measuring probe** to determine the **coordinates (X, Y, Z)** of points on a work-piece surface in a **3-D Cartesian system** . By measuring many points, **geometry** (dimensions, form, orientation, location) is computed by **software** . Called a **universal measuring machine** .

Bridge-type CMM (schematic)



Construction (main parts)

1. **Machine structure** : **bridge** , **cantilever** , **gantry** , **horizontal arm** , **column** types; made of **granite, aluminium, ceramic** - light, rigid, **low thermal expansion** ; **granite table (surface plate)** with T-slots / threaded inserts.
2. **Three mutually perpendicular axes X, Y, Z** guided on **air bearings** (frictionless, no wear); **drive** : motor / manual; **counterbalance** for Z axis.
3. **Measuring system (scales)** : **linear encoders / optical grating (Moire fringe), laser** on each axis - resolution **0.5 - 1 micron** ; some with **0.1 micron** .
4. **Probe system** : **touch-trigger probe** (kinematic seat with **three ball-rod contacts** ; the tip deflection breaks an electrical contact -> a **trigger signal** records the coordinates), **analog / scanning probe** (continuous displacement), **non-contact** (**laser, optical / video, white-light**), **probe head** (**indexable / motorized**), **stylus** (ruby ball tip, 0.3 - 8 mm) and **extensions** with **probe-changing rack** .
5. **Controller & computer** (**CNC**): motion control, data acquisition; **software** - alignment, geometry, **GD&T evaluation** , **CAD comparison** , **statistics** , reports.

Features / working

- **Measures** size, **form** (flatness, circularity, cylindricity), **position** , **orientation** , **profile** of **complex 3-D parts** in a **single set-up - versatile** ; **gears, cams, turbine**

blades, car bodies .

- Probe calibration with a reference sphere ; work coordinate system by alignment (plane-line-point, 3-2-1 method) - datum setting; auto-compensation for temperature and machine geometry errors (software error mapping).
- Operation modes : manual , motorized (joystick) , DCC (direct computer control; programmable CNC).
- Accuracy : typical volumetric accuracy = (1.5 + L/300) micron for L in mm; repeatability 0.5 micron .
- Advantages : flexibility , high accuracy , reduced set-up and inspection time , automatic data processing , no special gauges , operator independence , integration with CAD/CAM (reverse engineering).
- Limitations / error sources : cost , temperature & vibration , probe errors (lobing, pre-travel) , guideway geometry errors (21 errors for 3 axes), operator skill in programming , slow compared to gauging for mass production.

CMM type	Features / use
Bridge	most common , high accuracy, medium-large parts; fixed table
Cantilever	open access on three sides; small parts, good for sheet-metal ; less rigid
Gantry	very large parts (aircraft, car bodies); floor-mounted
Horizontal arm	car body inspection; large flat parts, dual-arm version
Portable arm / laser tracker	large assemblies, on-site; lower accuracy

Quick Revision / Important Questions

1. Elements of a screw thread; errors in threads. Two-wire & three-wire methods with formulae.
2. Floating carriage micrometer; thread gauges (plug, ring, snap); Taylor's principle for threads.
3. Gear tooth terminology; measurement of run-out, pitch, lead, backlash, profile.
4. Gear tooth vernier: chordal thickness; constant chord; base tangent method (examples).
5. CMM: construction, probes, features, advantages and limitations.